

C++ Programming

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Agenda

- class and object
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- Types of Member Functions
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- Constructor initializer list
- Exception Handling



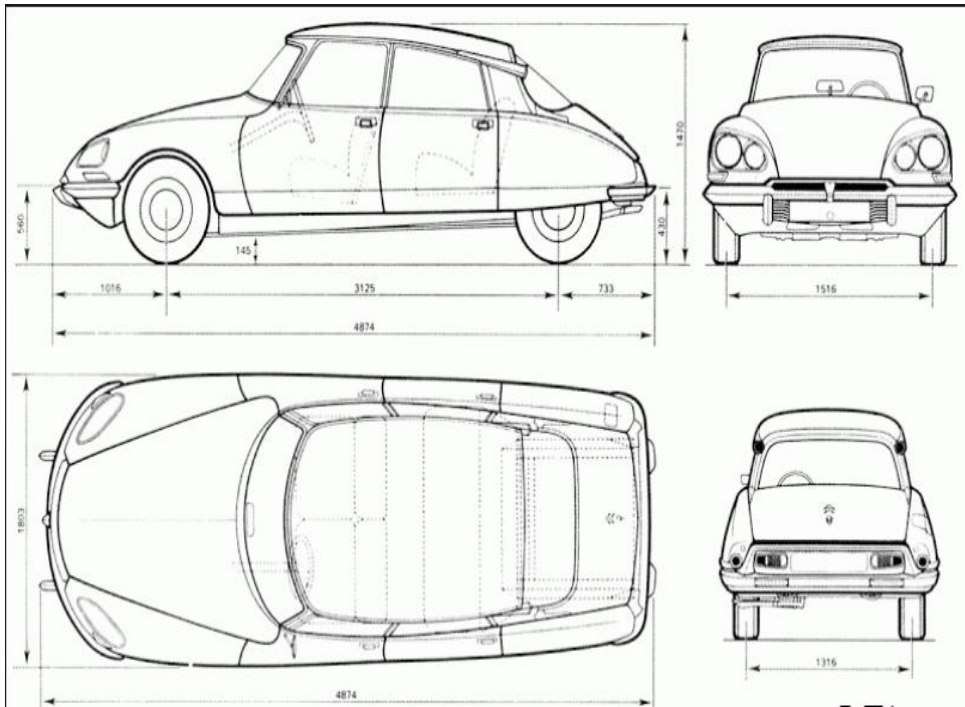
class and object

- Everything in C++ is associated with **classes and objects**, along with its properties and behavior.
- The class is **template** for **an Object**
- Or you may say it is a **blue print for an object**, by looking towards which an object gets instantiated.
- Class is collection of data member and member function.
- Object is an **instance** of a class.
- Entity that has **physical existence, can store data, send and receive message** to communicate with other objects.



Example of Class and Object

Blue Print of Car



Objects of a
Class Car



class and object

- Example:

Blue Print of building



Real life entity(Object)



Object

- An entity, which **get space inside memory** is called object.
- Object is used to access **data members** and **member function** of the class
- Process of creating object from a class is called **instantiation**
- **Object has**
 - Data members (***state*** of object)
 - Value stored inside object is called **state of the object**.
 - Value of data member represent **state of the object**.
 - Member function (***behavior*** of object)
 - Set of operation that we perform on object is called behaviour of an object.
 - Member function of class represent behaviour of the object.
 - is how object acts & reacts, when its state is changed & operations are done
 - Operations performed are also known as messages



Object

- **State:** State of an object contains all the **properties** of the object and **values of** each and every the **properties**.
- **Value Stored in side an object** is called state of an object.
- **Behavior:** Behavior is how an object **act and react**, when state of an object changes and function gets called.
- **Member function** of a class is nothing but the **behavior** of an object.
- **Identity:** Identity is **the property of a class** by which it gets **differentiated** from all other objects.



Data Member & Member Functions

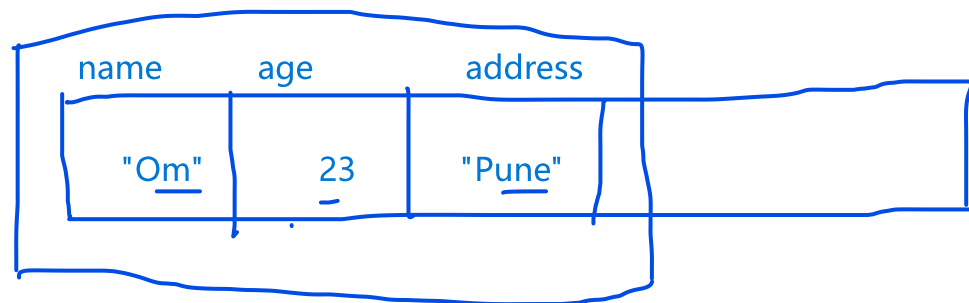
- Unique address(***identity*** of object)
 - Value of any data member, which is used to identify object uniquely is called its identity.
 - If state of object is same then its **address** can be considered as its identity.
- **Data members are the data variables and member functions are the functions used to manipulate these variables** and together these data members and member functions defines the properties and behavior of the objects in a Class.
- The data members and member functions of class can be accessed using the dot('.')



Types of Member Functions

- Constructor
- Facilitator
- Inspector
- Mutator
- Destructor
- **Constructor:**
 - Constructor is a **special member** function of a class **having same name** as a class and having no return type.
 - Constructor is used to **initialize an object/data member** of a class.
 - In entire **life cycle** of an object constructor **gets called only once**.

Person p1;



P1



Types of Constructor

- **Types of Constructor:**
 - 1. Parameterless Constructor
 - 2. Parameterised Constructor
 - 3. Default Constructor
- **1. Parameterless Constructor:** user defined default constructor
 - Parameterless constructor is a constructor which **does not have parameter** is called as parameterless constructor.
 - It is also called as **zero parameter constructor** of user defined default constructor.
- **2. Parameterised Constructor:**
 - Parameterised constructor is a constructor which **dose have one or more parameter** is called as parameterless constructor.
Parameterised
- **3. Default Constructor:**
 - If no constructors are explicitly declared in the class, a default constructor is **provided automatically by the compiler**



Types of Member Functions

- Facilitator:
 - Cause an object to perform some action or service.
- **Inspector**: getter
 - public member functions in a class that get (accessors) the value of data member of a class.
- **Mutator**:
 - a mutator method is a method used to **change the value** of a data member of the class.



Types of Member Functions

- **Destructor:**

- Destructor is also a special member function like constructor.
- An operation that **deinitializes an object**.
- A member function that **is invoked automatically** when the **object goes out of scope or is** explicitly destroyed by a call to **delete**.
- Destructor neither **requires any argument nor returns any value**.

```
~Person()  
{  
  
}
```

```
void abc()  
{  
    Person p1;  
    p1.display();  
}
```



This Pointer

- this is a **key word** in C++
- It is **implicit constant pointer** available in side the each non static member function of a class.
- This pointer stores the address of current object of a class.
- If **name of data member** and **parameter passed to the member function** is **same** then it becomes **mandatory** to use **this pointer** inside that function to differentiate between the data member and local parameter to the function.
- Following are the function which doesn't get this pointer
 - Global Function
 - Static Function
 - Friend Function

```
const int num1=10;  
num1=20;//not ok
```



Constructor's member initializer list

- If we want to initialize data members according to users requirement then we should use constructor body.

```
class Test
{
private:
    int num1;
    int num2;
    int num3;

public:
    Test( void )
    {
        this->num1 = 10;
        this->num2 = 20;
        this->num3 = num2;
    }
};
```

- If we want to initialize data member according to order of data member declaration then we can use **constructors member initializer list**.

```
class Test
{
private:
    int num1;
    int num2;
    int num3;

public:
    Test( void ) : num1( 10 ), num2( 20 ), num3( num2 )
    { }

};
```

Except array we can initialize any member inside constructors member initializer list.



Empty Class

Empty Class :-

A class which **do not have any Data member or member function** is called as the Empty class.

- We **can create an object** of a empty class.
- **Size** of an object of a empty class is **1 byte**.



Difference between class and Structure in C++

Class

Members of a class are private by default.

Member classes/structures of a class are private by default.

It is declared using the **class** keyword.

It is normally used for data abstraction and further inheritance.

Structure

Members of a structure are public by default.

Member classes/structures of a structure are public by default.

It is declared using the **struct** keyword.

It is normally used for the grouping of data



Exception Handling

- One of the **advantages of C++ over C** is Exception Handling. Exceptions are **runtime anomalies/error** or **abnormal conditions** that a program encounters during its execution.
- Exception is an **event or object** which is thrown at runtime.
- **Exception Handling** in C++ is a **process to handle runtime errors**. We perform exception handling so the **normal flow** of the application can be **maintained even after runtime errors**.
- Exception need to be handled other wise it would terminate the program abnormally.



Exception Handling:

- To create an application, most of the times we take help of operating system resources. Following are some the resources we use frequently.
 - 1. Memory
 - 2. File
 - 3. Socket etc.
- In C++ we can handle exceptions with the help of following keywords:
 - 1.try
 - 2.catch
 - 3.throw



Exception Handling:

- 1. try block/ try handler:

- if we want to **inspect** group of statements for **exception** then we should put such statement **inside try block** or **try handler**.

- Syntax:

```
try{
```

```
    code statement 1;
```

```
    code statement 2;
```

```
    code statement 3;
```

```
}
```

} Code which may possibly generate the exception



Exception Handling:

- **2. catch block/catch handler:**

- If we want to **use try** block then it is **necessary** to provide at **least one catch** block.
- To **handle exception** thrown **from try block** we should **use catch block** or catch handler. Single try block may have multiple catch block but it must have at least one catch block.

- **Syntax:**

```
try{  
    code statement 1;  
    code statement 2;  
    .....  
    code statement n;  
}  
catch()  
{  
    //Block of code to handle the error  
}
```

//Code which may possibly generate the exception



Exception Handling

- **Generic Catch block/Generic catch handler:**

- A catch block which can handle all types of exception is called generic catch block.
- In case if you don't have an idea which error will be thrown by the block of code then in such case you need to give generic block for the try.

- **Syntax:**

```
try{  
    code statement 1;  
    code statement 2;  
    .....  
    code statement n;  
}  
catch(...)  
{  
    //Block of code to handle the error  
}
```

} //Code which may possibly generate the unknown exception



Exception Handling

- **3. throw:** A program throws an exception when a problem shows up. This is done using a **throw** keyword.
- Example:

```
try
{
    cout<<"Before throw...!";
    throw 10;
    cout<<"After throw...!"
}
Catch(int n)
{
    cout<<"Error code:"<<n;
}
```



Thank You

